Learning-Based Tactile Posterior-Guided Search for Robotic Hex-Socket Engagement

Abstract— **Hex-socket unscrewing starts with one small move: the driver must enter the socket before torque can be applied. Visual alignment brings the tool close to the screw head, but the residual error can still match the chamfer capture range. An outward spiral search is safe against this error, but it keeps checking locations that contact has already ruled out. A single learned move covers less ground, but it is unsafe when touch shows direction better than it shows distance. This paper builds a search that uses both ideas together. A low-force conical probe records axial force, bending torque, and tip height while it moves. A trained ensemble turns this signal into a direction estimate and a contact-confidence estimate. These two estimates reweight a CAD prior over the socket center. The confident part of that estimate becomes a short, capped move, and the rest goes to a small local search. Capture, recovery, force limits, and the final check all run through one compliant control loop. In paired M8 simulations, this method keeps the same engagement rate as outward search, with a shorter path and less time. Tests on M10 and M12 sockets, and trials on a KUKA LBR iiwa, show the same method still works at other sizes. Learned touch, in short, is useful here mainly for choosing where to search, not for deciding that the driver has already engaged.**

Index Terms—Learning-based control, tactile sensing, robotic unscrewing, active perception, contact-rich manipulation, compliant interaction.

I. Introduction

Hex-socket unscrewing is usually treated as a torque-control problem. But many trials are actually decided before the screw is even turned. The driver first has to enter a recessed socket, and the clearance there is small compared with the residual error of a camera–CAD estimate [1]. Going down too early lands on the screw head. Searching from scratch is safer, but it wastes contact travel on locations that the first touch has already ruled out. This is a familiar part-mating problem [2], [3]. Here it comes with one more constraint: the search must finish inside a short manipulation cycle.

Compliance is the usual first defense against this kind of error. Remote-center compliance and impedance control let the mating geometry fix small misalignments on its own [2], [4]. When the offset is larger, though, the controller also has to decide where to touch next. Belief-space planning has been used for this [5]–[7]. Contact-implicit and complementarity methods instead treat the contact mode as part of the control problem [8]–[10]. Fast approximations of contact dynamics give another path to tractable planning [11]–[13]. Over the last few millimeters above a socket, though, the question is simpler: what motion is actually justified by the contact just felt?

Touch alone can answer part of this question, even without a camera on the contact [14]. Contact-pose estimation and active extrinsic sensing pull compact state estimates out of a prescribed motion [15], [16]. Learned tactile and visuotactile methods push this further, to richer sensors and less structured interactions [17]–[27]. These works show that touch carries real geometric information. None of them, though, says how far a robot should actually move when that information is still incomplete.

This gap matters for the conical probe used here. Its wrench pattern usually points at the bearing of the socket center. But it is much less sure about the radial offset. Treat the bearing as a point estimate, and the driver can overshoot a small socket. Ignore it, and the only useful touch made so far is wasted. Meta-learned and offline insertion policies address related adaptation problems [28], [29], and diffusion policies give strong general visuomotor priors [30]. Final engagement still needs explicit force and timing rules [31]–[33]. Differently, in this work, learning only picks where to search. It never gets to decide that the driver has engaged.

Figure 1 shows the setup and the coordinate convention used below. The interaction frame sits on the force-sensing tool. The fixture frame marks the flat socket geometry. The robot-base frame gives the commanded Cartesian reference. All z axes point up, so engagement depth grows along $-z_p$.

The conical probe gives a circular estimate over direction and contact state. It reweights a rough planar estimate instead of pinning down one socket center. The robot moves only as far as a safe, supported quantile, then searches the rest locally. Wrench limits, recovery, and the final check stay the same no matter what the model predicts. A shaky tactile estimate can shorten the route. It can never sign off on its own success.

The main contributions are as follows:

1) extracting a phase-indexed tactile signal that gives direction and contact support from one low-force conical touch, while keeping the direction–distance ambiguity of the socket center in plain view;
2) turning that signal into action: a risk-bounded shortcut plus a finite local search, both run through one shared compliant capture and recovery controller;
3) testing this in a paired study that checks engagement, task-level efficiency, direct-ray accuracy, recovery, and transfer across sizes, and then running it on real hardware.

The study stops at verified socket engagement. Turning the screw, releasing the thread, and extracting it are not part of this work.

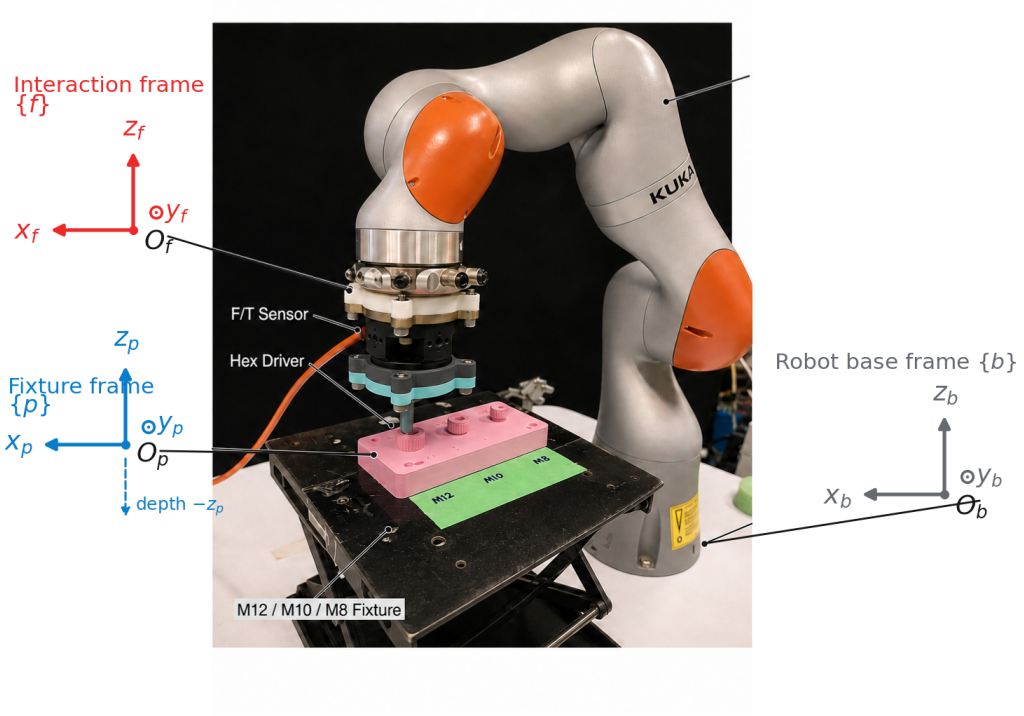


Fig. 1. Experimental platform and coordinate-frame convention. Frame $\{f\}$ sits on the force–torque-sensor/tool stack. Frame $\{p\}$ sits on the M12/M10/M8 socket plate. Frame $\{b\}$ sits on the KUKA LBR iiwa. All z axes point up. Engagement depth is measured along $-z_p$. The $\odot$ marks show y axes pointing out of the page.

II. Problem Formulation

A. Task and observations

Let $\{b\}$, $\{p\}$, and $\{f\}$ be the robot-base, fixture, and interaction frames in Fig. 1. All planar quantities below are written in $\{p\}$. Here $C \in \mathbb{R}^2$ is the unknown socket center. $x_0 \in \mathbb{R}^2$ is the tool position at the first stable top contact. $p_0(C)$ is a truncated visual/CAD prior over a finite set G. During interaction, the controller reads robot pose and the wrench

$$w = [F_x, F_y, F_z, \tau_x, \tau_y, \tau_z]^\top. \tag{1}$$

The true center and contact labels from simulation are held back, for offline checking only.

From the stable top contact, the tool runs a 15° low-force conical motion. With phase ϕ, the observation is resampled into

$$y(\phi) = \begin{bmatrix} F_z(\phi) & \tau_x(\phi) & \tau_y(\phi) & z_{\text{tip}}(\phi) \end{bmatrix}^\top, \quad \phi \in [0, 2\pi). \tag{2}$$

Note that probe phase ϕ is not the planar direction angle θ from x_0 to C. Those are two different things. The height channel records what comes with the wrench signal. It is not an insertion detector. For each channel c, the samples are ordered on the circle and linearly interpolated with periodic boundary conditions,

$$\begin{aligned} \phi_n &= \frac{2\pi n}{N_\phi}, \qquad n = 0, \dots, N_\phi - 1, \\ y_c[n] &= \mathcal{I}_{\mathbb{S}^1}\{y_c\}(\phi_n), \qquad N_\phi = 360. \end{aligned} \tag{3}$$

Here $\mathcal{I}_{\mathbb{S}^1}$ is the periodic interpolant. This step only fixes the input format. It does not smooth the raw signal.

A trial counts as successful only if sensor-side capture is followed by a steady bottom-mode dwell before $T_{\max} = 130$ s, and only if no force-limit stop happens first. This verified-engagement event is the main outcome we track. We also report completion time T_{eng}, measured planar path L_{xy}, and a contact-work proxy E_w, all as conditional efficiency measures. Peak wrench and the exit reason are kept too, to help read failures. Let γ_{cap}, γ_{bot}, and γ_{safe} be the capture, bottom-mode, and safety predicates. The success event is then:

$$\begin{aligned} \gamma_{\text{dwell}}(t) &= \mathbf{1}\left[\int_{t-\Delta_b}^{t} \gamma_{\text{bot}}(s)\, ds = \Delta_b\right], \\ \gamma_{\text{safe}}^{[0,t]} &= \min_{s \in [0,t]} \gamma_{\text{safe}}(s), \\ \mathcal{V} &= \{\exists\, t_c < t_b \le T_{\max} : \\ &\qquad \gamma_{\text{cap}}(t_c)\gamma_{\text{dwell}}(t_b)\gamma_{\text{safe}}^{[0,t_b]} = 1\}. \end{aligned} \tag{4}$$

For a verified trial that starts at stable touchdown t_0, we report

$$\begin{aligned} T_{\text{eng}} &= t_b - t_0, \qquad L_{xy} = \int_{t_0}^{t_b} \|\dot{x}(t)\|_2\, dt, \\ E_w &= \int_{t_0}^{t_b} \left(|F_{xy}^\top \dot{x}| + |\tau^\top \omega|\right) dt. \end{aligned} \tag{5}$$

B. Direction–distance ambiguity

Let $r = \|C - x_0\|$. Let the circular distance be

$$d_{S^1}(a, b) = |\text{atan2}(\sin(a-b), \cos(a-b))| \in [0, \pi]. \quad (6)$$

Then $\epsilon = d_{S^1}(\hat{\theta}, \theta^\star)$ is the circular error of a direct ray. Under ideal planar motion, the closest gap between that ray and the socket center is

$$d_\perp = r|\sin\epsilon|, \qquad d_\parallel = r\cos\epsilon. \quad (7)$$

Here r_{eff} is a safe planar capture radius. It is fixed per socket size from the hex clearance, the chamfer shape, and how much slack the compliant contact allows. The ray only hits the capture disk if $d_\perp \le r_{\text{eff}}$ and $d_\parallel > 0$. So a small angle error is harmless near the center. Far from the center, the same small error can miss completely. The conical scan never sees r, so a direction estimate alone cannot pick a safe endpoint. Differently, in this work, we use two steps instead: touch first updates a planar center belief, and only then does it decide how far to travel.

III. Tactile Posterior-Guided Control Framework

Figure 2 splits the system into two parts: learned tactile inference and belief-space routing on one side, physical interaction control on the other. Blue blocks pull contact-state and direction estimates out of the conical touch. Green blocks then mix that estimate with CAD support, and set a capped shortcut plus a residual search. A shortcut that misses is itself useful: it becomes negative evidence before the residual route runs. Below the authority line, the black loop drives the robot–socket contact. The red loop sits apart from it and keeps the recovery and terminal calls.

A. Circular tactile posterior

Three separately trained models map the 4×360 signal to logits over $K = 48$ direction bins. After temperature calibration, their average is

$$p_k = \frac{1}{M}\sum_{m=1}^{M} \text{softmax}\left(\frac{g_m(y)}{T_m}\right)_k, \qquad M = 3. \quad (8)$$

Its circular mean and concentration are

$$\mu = \arg\sum_k p_k e^{i\theta_k}, \qquad \kappa = \left|\sum_k p_k e^{i\theta_k}\right|,$$
$$\hat{d} = [\cos\mu, \sin\mu]^\top. \quad (9)$$

A second head gives a rough contact-support score $p(s)$. An adaptive prediction set [34] sums up angular support. Separately, the largest top-1 gap Δ_{ens} across ensemble members marks how much the models disagree. These are only descriptions of the observation. None of them counts as capture. More precisely, order the bins so that $p_{\pi(1)} \ge \cdots \ge p_{\pi(K)}$. At kept mass $1-\eta$, the set used is

$$r_\eta = \min\left\{r : \sum_{\ell=1}^{r} p_{\pi(\ell)} \ge 1-\eta\right\},$$
$$\mathcal{A}_\eta = \{\pi(1), \ldots, \pi(r_\eta)\}. \quad (10)$$

Let $\hat{\theta}_m = \theta_{\arg\max_k p_{m,k}}$ be the top bin of member m. The disagreement across the ensemble is then

$$\Delta_{\text{ens}} = \max_{m,n\in\{1,\ldots,M\}} d_{S^1}(\hat{\theta}_m, \hat{\theta}_n). \quad (11)$$

We also keep the entropy of the posterior, in nats, as one more check on how spread out it is,

$$H(p) = -\sum_{k=1}^{K} p_k \log p_k, \quad (12)$$

A tight posterior can still hide members that disagree. A spread-out posterior can still hide members that agree. These are not the same thing, so we keep both numbers. Figure 3 shows the whole chain. Real keyframes show the conical motion as run on hardware. One held-out MuJoCo record then shows the raw signal, the posterior, and the route it produces.

B. From tactile direction to center belief

The visual prior is a set of $N = 4096$ center particles,

$$C_j = x_0 + \delta_j, \qquad \delta_j \sim \mathcal{N}(0, \sigma_v^2 I) \mid \|\delta_j\| \le R_v. \quad (13)$$

For M8, $\sigma_v = 2$ mm and $R_v = 4.895$ mm. Each particle gets a new weight, from the posterior probability of its angular bin times a radial likelihood L_s tied to contact support:

$$\bar{w}_j = \frac{p_{b(\arg\delta_j)}\,[0.02 + 0.98\sum_s p(s)L_s(\|\delta_j\|)]}{\sum_\ell p_{b(\arg\delta_\ell)}\,[0.02 + 0.98\sum_s p(s)L_s(\|\delta_\ell\|)]}. \quad (14)$$

The small uniform term stops one shaky reading from erasing the geometric support. More to the point, it keeps the radial ambiguity in plain view, instead of burying it inside a point estimate. For later use, the weighted empirical quantile is

$$Q_\rho^w(a) = \inf\left\{u : \sum_{j=1}^{N} \bar{w}_j \mathbf{1}[a_j \le u] \ge \rho\right\}. \quad (15)$$

C. Risk-bounded shortcut and residual search

Project each particle onto the estimated ray, $z_j = (C_j - x_0)^\top \hat{d}$. Call $p_- = \sum_j \bar{w}_j \mathbf{1}[z_j \le 0]$ the posterior mass sitting behind the current contact. Observation quality is then

$$q = \text{clip}(\kappa[\cos(\Delta_{\text{ens}}/2)]_+(1-2p_-), 0, 1),$$
$$\alpha(q) = 0.10 + 0.15q, \qquad \lambda(q) = 0.35 + 0.30q. \quad (16)$$

So evidence that is spread out, disagreeing, or pointing backward earns less direct travel.

Let $Q_\rho^w(\cdot)$ be a weighted empirical quantile, and L_c an optional command cap. The allowed shortcut is then

$$L_{\text{proj}} = [Q^w_{\alpha(q)}(z) - 0.25 r_{\text{eff}}]_+,$$
$$L_{\text{rad}} = \lambda(q) Q^w_{0.99}(\|C - x_0\|),$$
$$L^* = \begin{cases} 0, & p_- \ge 0.20, \\ \min\{L_{\text{proj}}, L_{\text{rad}}, L_c\}, & p_- < 0.20, \end{cases}$$
$$x_s = x_0 + L^* \hat{d}. \quad (17)$$

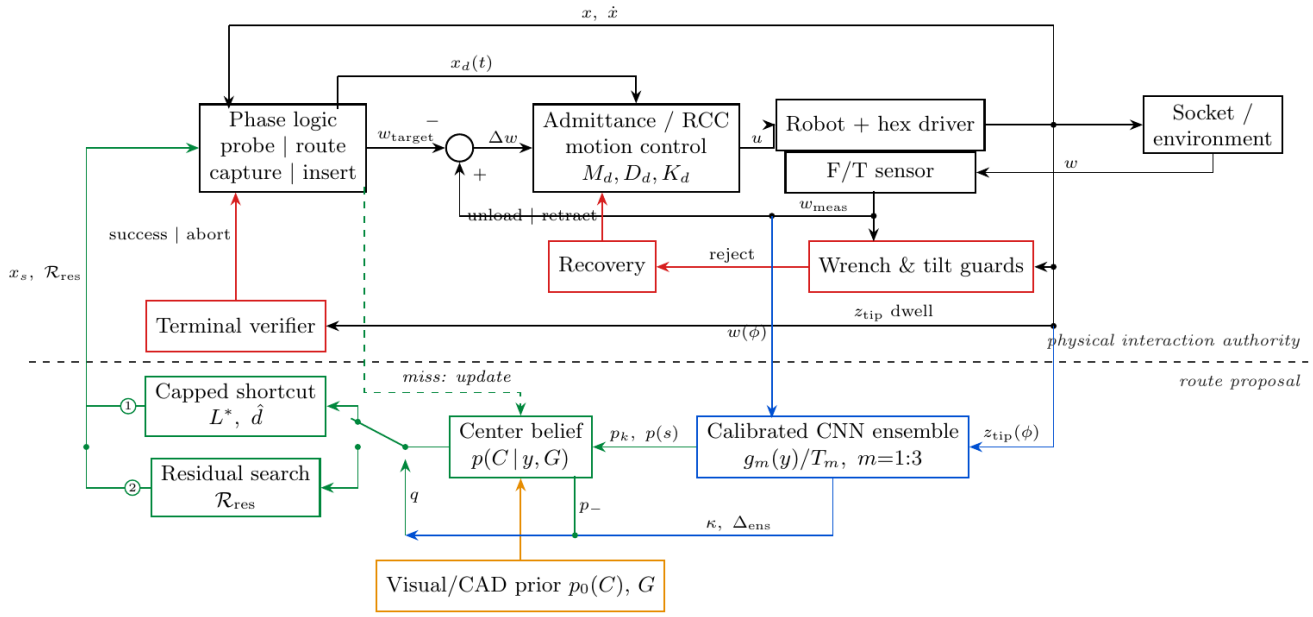


Fig. 2. Control architecture of the proposed framework. Blue blocks are learning-based tactile inference. Orange is the visual/CAD prior. Green is the center-belief and task-planning path. Black is the physical control loop. Red is the wrench, recovery, and terminal guards. The dashed line separates route proposals from physical control.

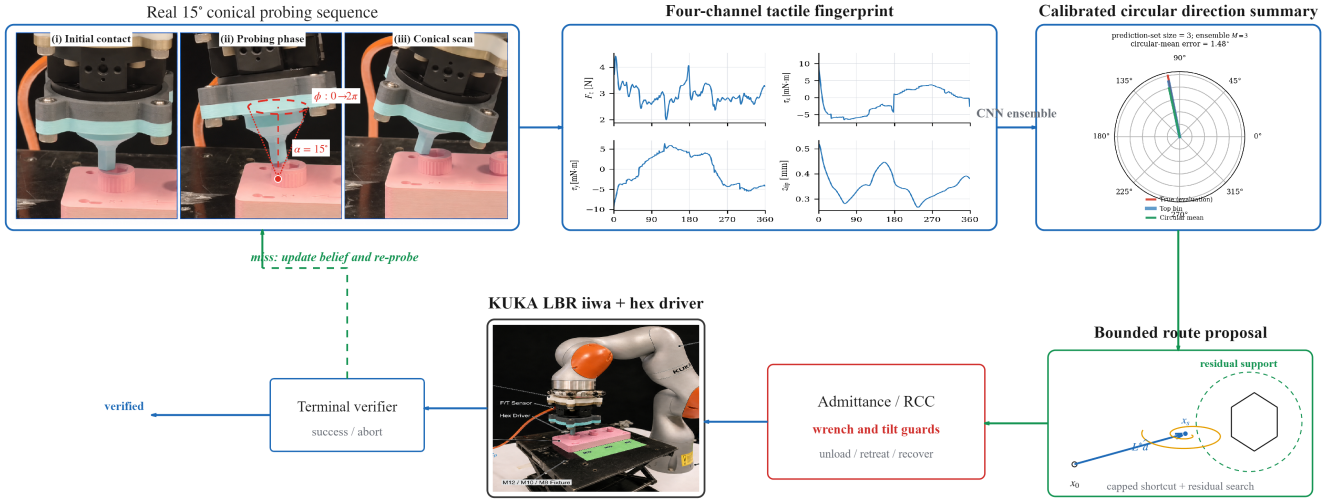


Fig. 3. Closed-loop tactile-guided search. Real keyframes show first contact, probing, and the 15° conical scan. A held-out MuJoCo example shows the four-channel signal and the circular summary built from it (three-bin prediction set, ensemble size $M = 3$, circular-mean error 1.48°). Direction and confidence set the shortcut and the residual route. Admittance/RCC, wrench and tilt guards, and the final check stay in charge the whole time. A miss updates the belief before the next probe.

Proposition 1 (Posterior mass retention): Assume $p_- < 0.20$, $L^* > 0$, and that the shortcut endpoint is reached exactly. For a fixed weighted-particle set, the shortcut obeys

$$\widehat{\mathbb{P}}_w[z \geq L^* + 0.25 r_{\text{eff}}] \geq 1 - \alpha(q),$$
$$L^* \leq \lambda(q) Q^w_{0.99}(\|C - x_0\|). \tag{18}$$

Proof: Take $L^* > 0$. Then $L^* \leq L_{\text{proj}}$, so $L^* + 0.25 r_{\text{eff}} \leq Q^w_{\alpha(q)}(z)$. By how the weighted quantile is built, at least $1 - \alpha(q)$ of the empirical mass sits at or above that point. That is the first bound. We also have $L^* \leq L_{\text{rad}}$. The second bound follows the same way. ■

This result only allocates posterior support. It does not promise capture. The residual search radius after the shortcut is

$$R_{\text{res}} = [Q^w_{0.99}(\|C - x_s\|) - r_{\text{eff}}]_+ . \tag{19}$$

Figure 4 shows this split. In the case shown, a clear reading allows 2.873 mm of direct travel and leaves a 1.732 mm local search region.

The measured shortcut is also a plain physical observation. For an executed move $a : x \to x_a$, let $d_a(C)$ be the shortest planar gap from C to that move. The runtime uses a smooth

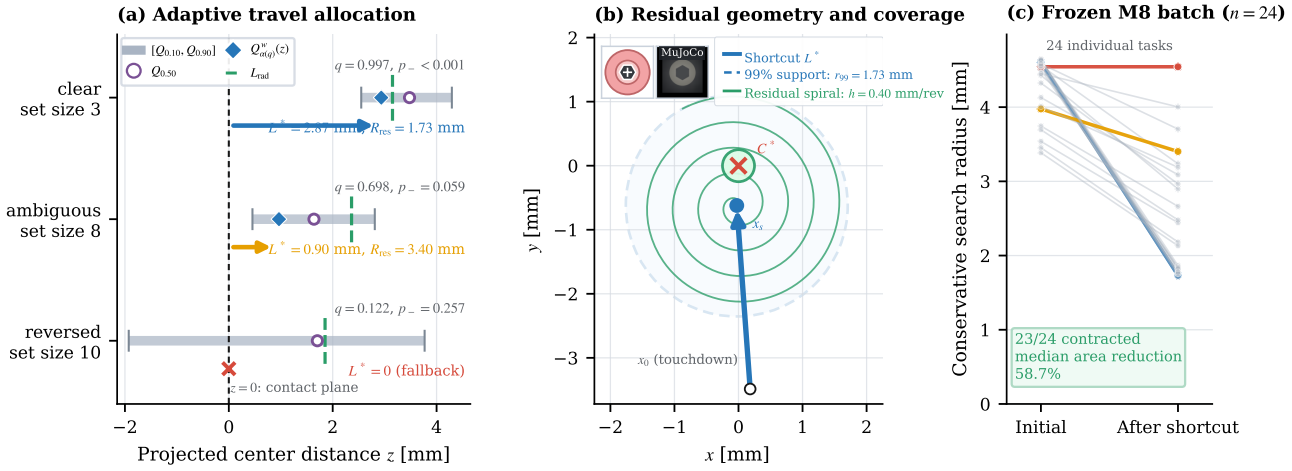


Fig. 4. Quality-adaptive posterior budgeting and residual search. (a) Three frozen M8 cases. Direction quality and reverse posterior mass set the allowed travel, including the $L^* = 0$ fallback. Prediction-set size is 3, 8, and 10 bins in these cases. (That is not the same number as the ensemble size, $M = 3$.) (b) The clear-observation command stops at a bounded endpoint (blue arrow). The dashed disk is the remaining weighted 99% radial envelope, and the green curve is the finite local spiral. The centered-socket and MuJoCo insets are not to scale. (c) How the conservative search disk shrinks, paired across all 24 frozen M8 tasks, with the three cases from (a) marked. Ground-truth center and capture-set marks are shown only for grading. The controller never sees them. All quantitative marks come from trial records.

capture likelihood,

$$d_a(C) = \min_{s\in[0,1]} \|C - [x + s(x_a - x)]\|_2,$$
$$\ell_{\text{cap}}(C \mid a) = \varepsilon_\ell + (1 - 2\varepsilon_\ell)\sigma\left(\frac{r_{\text{eff}} - d_a(C)}{h_\ell}\right), \tag{20}$$

Here h_ℓ sets the width of the contact boundary, and ε_ℓ keeps particles from dying out completely. When the measured endpoint is valid and still in guarded contact, but capture is rejected, weights update as

$$\bar{w}_j^+ = \frac{\bar{w}_j^-[1 - \ell_{\text{cap}}(C_j \mid a)]}{\sum_{\ell=1}^N \bar{w}_\ell^-[1 - \ell_{\text{cap}}(C_\ell \mid a)]}. \tag{21}$$

The residual geometry is then rebuilt around the measured endpoint. An endpoint that is invalid or never observed does not trigger this update.

The PGS-O variant covers the residual region with a finite local Archimedean spiral, pitch h_{sp},

$$x_{\text{sp}}(\psi) = x_s + \frac{h_{\text{sp}}\psi}{2\pi}\begin{bmatrix}\cos\psi & \sin\psi\end{bmatrix}^\top,$$
$$0 \le \psi \le \frac{2\pi R_{\text{res}}}{h_{\text{sp}}}. \tag{22}$$

The PGS-G variant instead scores each candidate move $a : x \to x_a$ by

$$U(a) = \beta_I I(C; O \mid a) + \beta_P \widehat{\mathbb{P}}_w([x, x_a] \cap B_{r_{\text{eff}}}(C) \neq \varnothing) - c_0 - c_\ell\|x_a - x\|. \tag{23}$$

The first term rewards a contact that tells the most. The second rewards a move that covers likely capture disks. The last terms simply tax switching and travel. When further contact stops being useful, the planner falls back to plain coverage. It does not force entropy to keep dropping. For the binary outcome $O \in \{\text{hit}, \text{miss}\}$, the information gain in (23) works out to

$$I(C; O \mid a) = H(\bar{w}) - \sum_{o\in\{\text{hit},\text{miss}\}} P(o \mid a) H(\bar{w} \mid o, a), \tag{24}$$

and the next move is picked only from what is left of the action set $\mathcal{A}_{\text{rem}}$:

$$a^* = \arg\max_{a\in\mathcal{A}_{\text{rem}}} U(a). \tag{25}$$

D. Compliant capture, recovery, and verification

Every search policy shares the same task-frame admittance/RCC controller,

$$M_d\ddot{\xi} + D_d\dot{\xi} + K_d\xi = w_{\text{meas}} - w_{\text{target}}, \tag{26}$$

plus rate saturation and a causal first-order wrench filter,

$$\alpha_f = \frac{\Delta t}{(2\pi f_c)^{-1} + \Delta t},$$
$$\bar{w}_t = \bar{w}_{t-1} + \alpha_f(w_t - \bar{w}_{t-1}). \tag{27}$$

Each wrench and tilt component is checked on its own,

$$\gamma_{\text{safe}}(t) = \mathbf{1}[\|\bar{w}(t) \oslash w_{\max}\|_\infty \le 1 \ \wedge\ |\alpha_{\text{tilt}}(t)| \le \alpha_{\max}], \tag{28}$$

where $\oslash$ is elementwise division. A capture cue only hands off to an insertion attempt. It does not force one. If progress stalls or tilt gets too large, the recovery operator kicks in:

$$\mathcal{R} = \text{retouch}_{0.20\,\text{s}} \circ \text{upright} \circ \text{retract}_{0.35\,\text{mm},\,0.15\,\text{s}} \circ \text{unload}_{0.10\,\text{s}}. \tag{29}$$

It picks up from the measured state. At most three recoveries are allowed. With axial progress $d(t) = z_{\text{tip}}(t_0) - z_{\text{tip}}(t)$,

the bottom and terminal predicates are

$$\gamma_{\mathrm{bot}}(t) = \mathbf{1}[d(t) \geq d_{\mathrm{req}} - \varepsilon_d],$$
$$\gamma_{\mathrm{term}}(t) = \gamma_{\mathrm{safe}}(t)\mathbf{1}\left[\int_{t-\Delta_b}^{t} \gamma_{\mathrm{bot}}(s)\, ds = \Delta_b\right]. \quad (30)$$

Here ε_d is the registered terminal-depth tolerance. Engagement is accepted only once this independent dwell holds. A force-limit hit, or a budget that runs out, ends the trial instead.

IV. Experiments

A. Setup, baselines, and measures

The main simulation benchmark has 24 task tuples, fixed before any policy is compared. A Sobol design varies planar offset, socket phase, sliding friction, contact dynamics, target force, sensing delay, and the posterior/noise seeds, all together. Radial offset spans 1.74–4.88 mm, sliding friction 0.564–0.945, contact time constant 40.6–69.7 ms, sensing delay 2.00–7.87 ms, and target downward force 2.50–3.44 N. Every policy gets the same tuple, the same 130 s budget, the same capture controller, and the same terminal check. No simulator reset and no budget extension is allowed.

Four task-level policies are compared. (i) PGS-O uses the posterior shortcut, then an outward residual spiral. (ii) PGS-G keeps the shortcut but swaps the residual for an entropy grid, which picks the next low-force contact from expected information gain and coverage. (iii) *Outward* runs a plain global Archimedean search, with no tactile shortcut at all. (iv) *Direct-4C* spends up to three physical rays along directions read off the four-channel signal. A missed direct ray unloads, shifts 0.25 mm across the ray, and repeats the full probe in the same rollout.

A second direct arm, *Direct-3C*, is used only for a representation check. It swaps $[F_z, \tau_x, \tau_y, z_{\mathrm{tip}}]$ for the three polar channels $[x_z, M_x, M_y]$ plus a 12-sector decision, following the contact-pose idea in [15]. This is our own proxy built for the same task. It is not a copy of that reference controller. Recent force-guided and belief-space exploration work [6], [7] matters for context. We do not give it a quantitative row, though, since its sensing, action, and terminal rules are not reproduced here.

Completion time stops at the verified terminal dwell. Planar path is the integral of measured tip motion. A force-limit stop, a protected post-capture stop, or running out of search budget all count as failure. Medians inside the successful group are only descriptive. Efficiency claims instead come from paired differences across joint completions, and binary recovery comparisons use exact paired tests. Every curve we show is rebuilt from frozen per-trial simulator records or timestamped hardware logs. No series is interpolated, smoothed, or filled in. The only changes made are the stated frame changes, same-run baseline subtraction, grouping, and cumulative counting.

B. Four-channel tactile representation

This check holds the first conical touch, the task tuple, the noise seed, the direct-ray budget, and the post-capture controller all fixed. So it asks a narrower question than the full policy comparison: does adding the height channel and the circular posterior change the direction the controller actually uses? Figure 5 shows the full paired set plus one audited example. Direct-4C has a smaller median error. Its tail is heavier, though, so we cannot call it uniformly more accurate.

TABLE I

Same-probe direct-ray ablation on 24 paired M8 starts. Exact McNemar values apply only to the binary paired outcomes.

Measure	Direct-4C	Direct-3C	Paired p
Median error	4.75°	7.41°	–
90th-percentile error	31.45°	13.79°	–
Within 15°	19/24	22/24	0.375
First-ray capture	18/24	16/24	0.6875
Any-ray capture, ≤ 3	19/24	19/24	1.0
Verified terminal	17/24	16/24	1.0

The result here is deliberately mixed. The fourth channel tightens the central error and adds two first-ray captures. The sectorized proxy, on the other hand, gets the smaller 90th-percentile error and three more cases within 15°. The any-ray capture count ties, and the terminal gap is only one trial (Table I). So we keep the four-channel signal because it feeds the shortcut and the entropy grid, not because this one batch proves it always points better.

C. Paired M8 closed-loop search

Figure 6 shows how the four policies use the same muted first contact in Sobol 016. (i) PGS-G spends the weak direction cue on a short move, then picks residual contacts from the entropy grid. (ii) PGS-O follows with an outward residual spiral. (iii) Outward pays for full coverage from the same starting contact. (iv) Direct-4C uses up its three physical rays. Frames are lined up by progress within each policy, so each row stays in order. What decides efficiency is the raw time and path, not how the frames happen to be spaced on the page.

Figure 7 and Table II cover the full paired set. The question the curves answer is simple: given a fixed time or a fixed distance, how many paired tasks have already engaged? PGS-O gets there in 20/24 cases. PGS-G and Outward tie at 19/24. Direct-4C often finishes fast, when its estimated ray happens to land in the capture region, but it settles at a lower total.

D. Efficiency and component ablations

PGS-O and Outward both finish 19 trials. On that shared subset, PGS-O covers less ground in 19 of 19, and takes less time in 18 of 19. Median paired gaps are -41.14 mm and -16.79 s, with exact sign-test values $p = 3.8 \times 10^{-6}$ and $p = 7.6 \times 10^{-5}$. This test compares the full probe–shortcut–residual policy against plain global coverage. On its own, it does not pull the shortcut apart from every other choice in the policy.

PGS-G and PGS-O share 19 completions. Median paired path gap: -0.061 mm (11/19 shorter, Wilcoxon $p = 0.465$).

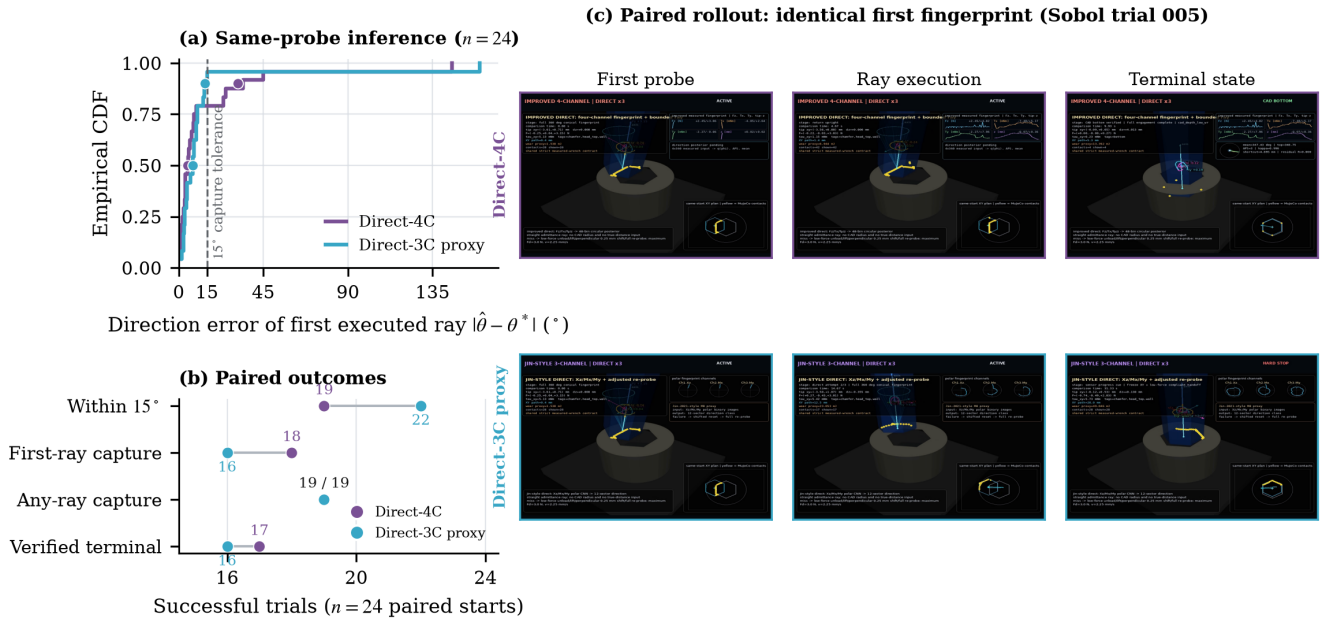


Fig. 5. Controlled tactile-representation ablation. (a) Empirical CDF of the first direction the controller actually runs, on all 24 paired M8 starts. Circles mark the median and the 90th percentile. (b) Paired capture and terminal counts. (c) Frames from Sobol 005, same raw fingerprint for both methods. Direct-4C captures on its first ray and reaches the terminal set. The three-channel proxy instead pays for three full probes, hands off to the shared capture controller on the third try, and then exits at a wrench limit. Direct-4C is the proposed representation. Direct-3C is the three-channel proxy. Panel (c) is one case, not a ranking across the population.

TABLE II

M8 ENGAGEMENT RESULTS ON 24 PAIRED STARTS. T_{eng}, L_{xy}, AND E_w ARE MEDIANS OVER SUCCESSFUL TRIALS. $\|F\|_{\max}$ IS THE MEDIAN TRIAL MAXIMUM OVER ALL STARTS. THE LAST COLUMN LISTS HARD-STOP, PROTECTED, AND BUDGET/RADIUS EXITS ($H/P/B$).

Method	Verified	T_{eng} [s]	L_{xy} [mm]	E_w [mJ]	$\|F\|_{\max}$ [N]	$H/P/B$
PGS-O	**20/24**	13.87	14.52	28.48	7.33	3/0/1
PGS-G	19/24	13.74	6.76	9.54	7.69	4/0/1
Outward	19/24	34.63	71.27	120.38	7.45	4/1/0
Direct-4C	17/24	9.22	6.07	13.67	7.33	3/0/4

Median paired time gap: $+0.536$ s (7/19 faster, $p = 0.196$). In other words, the gain over Outward cannot be pinned on the entropy grid. In this benchmark, the residual disks are only a few millimeters wide, so a local spiral already covers them with little wasted motion. The grid also scores contacts with the same rough likelihood used for the belief update, and close candidate scores, model error, plus the cost of unloading and touching again, can easily erase a small geometric gain. Bigger search regions or a better-calibrated likelihood might change this. The 24 starts here do not show that advantage.

Recovery is checked on a separate, fixed 12-case post-capture stress set. With the RCC and wrench guards left alone, turning recovery off gives no verified engagement. Allowing up to three recoveries gives 9/12, with no case going the other way (exact two-sided $p = 0.00390625$). Two force-limit exits and one protected exit remain. Recovery changes what state is reachable after a rejection. It does not force a win.

TABLE III

CROSS-SIZE ENGAGEMENT ($n = 12$ PER CELL): VERIFIED / FORCE-LIMIT / PROTECTED TERMINATION. REMAINING TRIALS END BY BUDGET OR TERMINAL REJECTION.

Interface	Dev M10	Dev M12	Test M10	Test M12
M8 zero-shot	12/0/0	10/1/1	12/0/0	10/0/1
Calibration only	12/0/0	10/0/1	12/0/0	11/0/0
Anchored fine-tuning	12/0/0	10/0/2	10/1/1	10/0/2
Privileged center	12/0/0	11/0/1	12/0/0	12/0/0

E. *Cross-size simulation*

We next check whether this narrow learned interface transfers from M8 to M10/M12, keeping the physical controller fixed. For each size, 72 training, 36 validation, and 36 calibration scans are split by sixfold symmetry. The encoder stays fixed. Only the contact-state and direction heads get updated, with an ℓ_2 anchor back to the M8 weights. A concentration and source-agreement gate turns off the shortcut when the adapted head looks unsure. Residual search and every physical guard stay untouched.

Table III shows no steady held-out gain from anchored



Fig. 6. Chronological MuJoCo keyframes, one muted-fingerprint M8 case (Sobol 016). Rows: PGS-G, PGS-O, Outward, Direct-4C. Columns: early probe, route/residual search, terminal state. Yellow points mark measured contacts. Each inset is the task-plane trace from that same start. PGS-G, PGS-O, and Outward all reach CAD-bottom verification. Direct-4C runs out of its ray budget first, with no capture. Badges show values from this one frozen trial. They will not match the medians in Table II.

fine-tuning over zero-shot or calibration-only transfer, even though it looks good on development data. So we keep residual search and the contact guards outside the transferred network, and fit the observation model again once it is on real hardware.

F. Hardware evaluation

Hardware data came from the tool and fixture registered in place. For each of M8, M10, and M12, the campaign has 16 clean conical scans, four development holdouts, four quarantined points, and two repeated sentinels. All 78 scheduled traces passed the acquisition checks, but their phase-aligned signals looked clearly different from the rendered ones. Coulomb and stiction error, deformed printed parts, sensor bias, fixture/TCP registration, and the gravity moment at the Bota sensor all add up to this gap. So the circular model used on the robot was fit again, on signals from the installed stack. These records carry the full hardware effect. They do not fix session drift, wear, temperature changes, or shifts across socket size. The M12 holdouts were already seen during development, so we do not treat them as an untouched test set.

Selected M12 runs went through the whole sequence: guarded touchdown, conical scan, bounded routing, capture re-check, staged insertion, and terminal dwell. Named PGS-O, PGS-G, and Direct-4C runs all reached the terminal event. Outward went over the 130 s budget. When the same run was let continue past that limit, it engaged after about 395 s. Two fixed-site M10 PGS-G pilots also reached terminal depths of 4.834 and 4.726 mm.

Figure 8 shows one named M12 PGS-G run, start to finish, with no mixing of trials. Its 240-sample real signal gave a 3.00 mm shortcut. The measured endpoint missed the command by 0.054 mm, and did not capture. That negative result then redirected the residual policy. Capture came 42.45 s after touchdown. The terminal candidate was logged at 116.63 s, with 5.792 mm of measured insertion depth. These numbers describe only that one run, not a general rate.

Figure 9 lines up two runs from the same starting point.

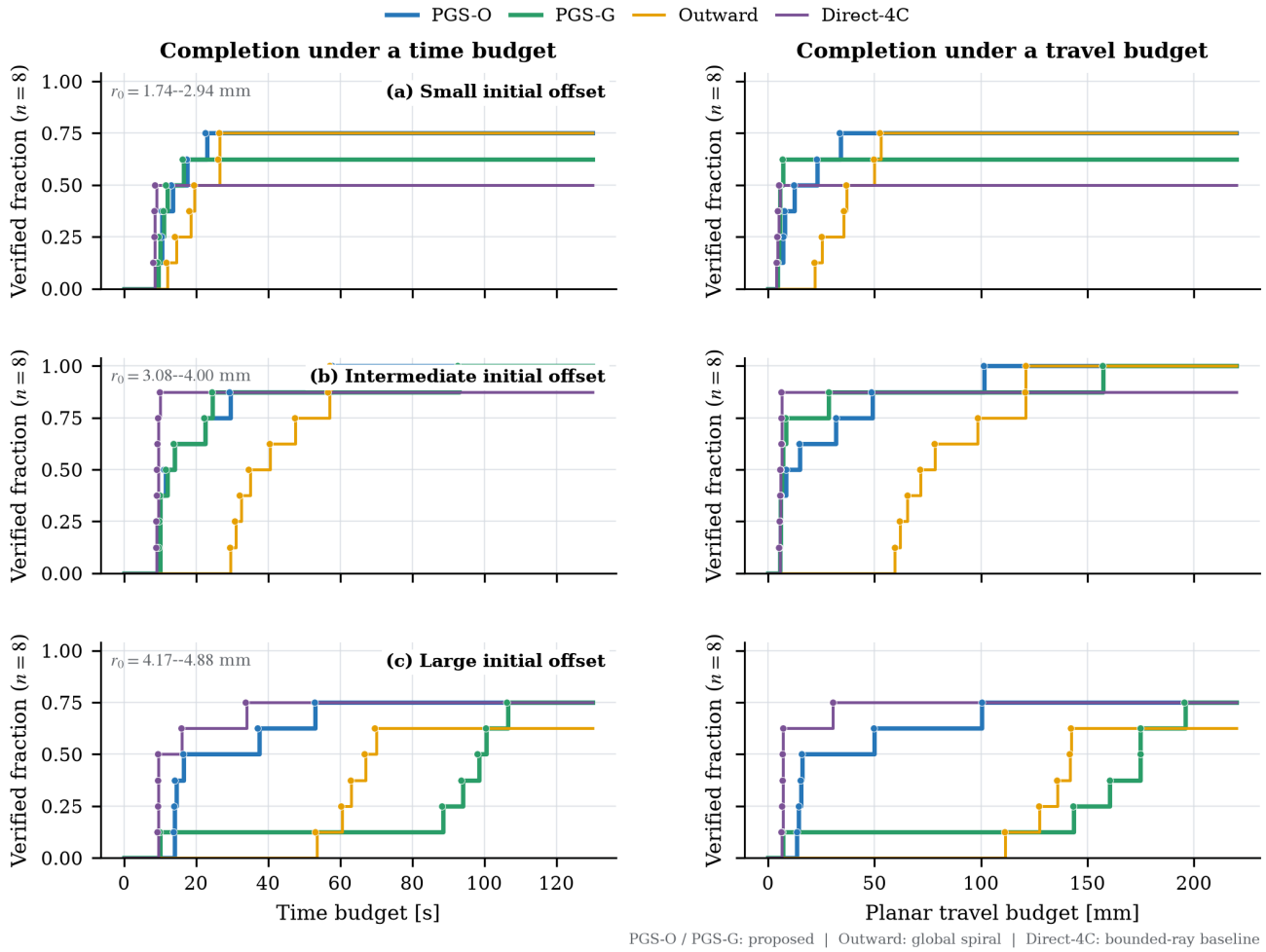


Fig. 7. Budgeted M8 engagement, 24 paired starts, split into three equal-size offset groups: (a) small, (b) medium, (c) large. Left column: fraction of eight tasks done within a time budget. Right column: same, using measured planar travel as the budget. Each row has $n = 8$ trials. The ordinate is the fraction that has reached verified engagement. Step points mark recorded terminal times and travel distances. Nothing here is smoothed or interpolated. A failed trial adds no step, so each plateau reflects both success and speed.

These are only mechanism checks. Two runs cannot give a success rate. PGS-G rejects the endpoint of its bounded shortcut and clears the rest of the route after one conical touch. Direct-4C misses on its first ray, probes again, and then captures. Both eventually reach the terminal state, under an extended time limit, so their paths, loads, and event logs can all be checked on the real robot.

Figure 10 lines up the full chain, from posterior to contact, for one fixed-seed four-policy M12 run. Every column uses the same registered offset and the same plotting scale. All three tactile methods pick the same main direction here, but they split up the rest of the route differently. PGS-G uses its negative evidence to replan a tight grid route. PGS-O covers the remaining region with a local spiral. Direct-4C commits to one bounded ray. Outward ignores the posterior and simply expands a global spiral. The times and paths shown belong only to these chosen runs. They are not an estimate of policy-level success or speed.

Table IV lists the hardware records used in this paper. The scan campaign checks acquisition across all three socket sizes. The M12 method runs and the two M10 pilots exercise the closed loop. These runs came from named or fixed starts, so the table reports what was seen, not a success rate.

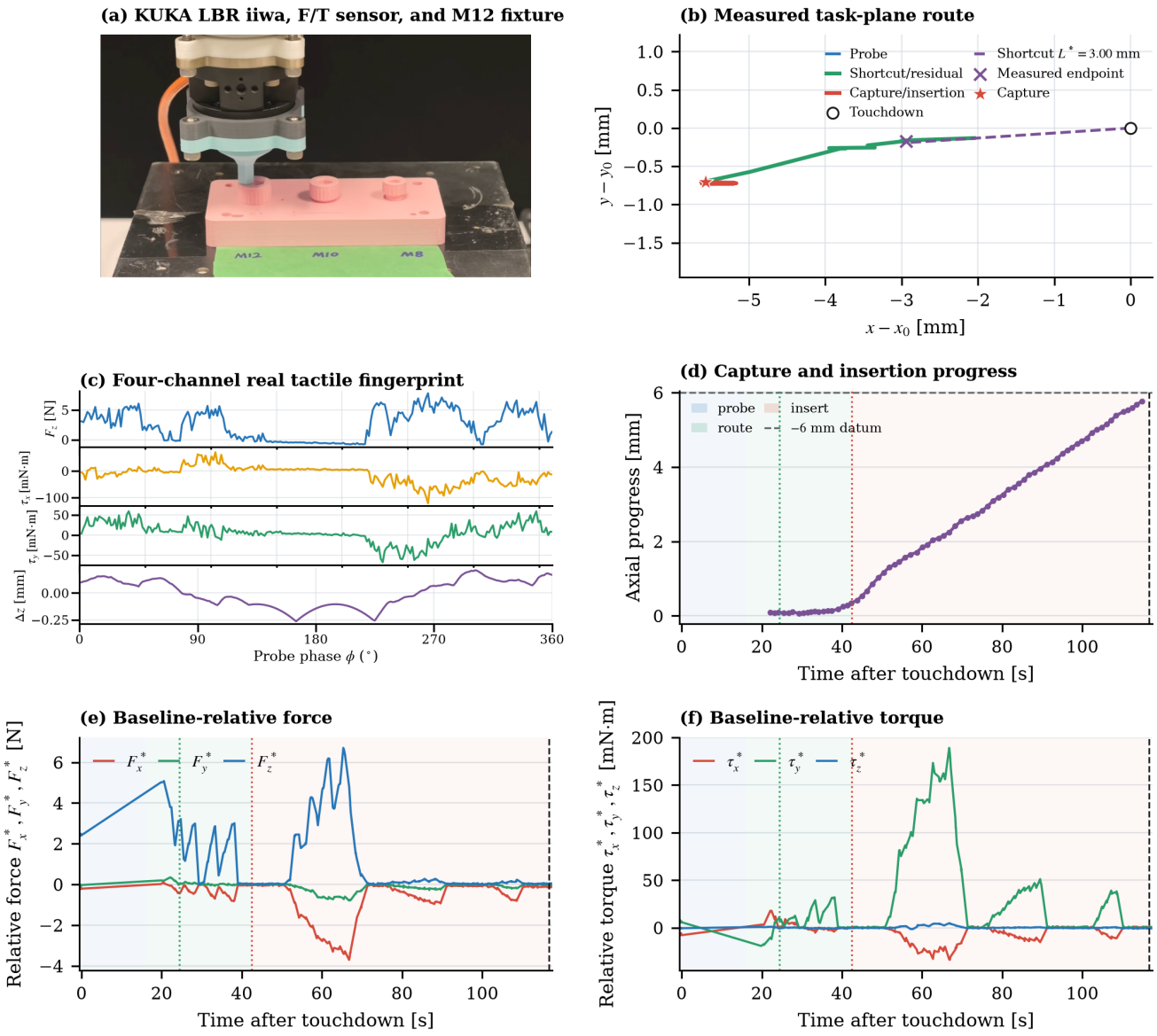


Fig. 8. Same-run hardware evidence, one M12 PGS-G run. (a) KUKA LBR iiwa, Bota F/T sensor, hex driver, multisize fixture. (b) Measured task-plane route, with the commanded shortcut, the measured endpoint, and the confirmed capture. (c) The 240-sample four-channel signal; tip height is centered on its scan mean for display. (d) Axial progress from the touchdown height, with the −6 mm reference line. (e),(f) Baseline-relative force and torque at the Bota origin. Blue, green, and red shading mark probe, route, and insertion. Dotted lines mark the negative evidence and the confirmed capture. The black dashed line marks the terminal event. Every curve here is a raw, unsmoothed timestamped record. The wrench has not been moved to the tool tip or corrected against a certified gravity model.

TABLE IV

HARDWARE EVIDENCE USED IN THE PAPER. NAMED AND FIXED-START EXECUTIONS SERVE AS MECHANISM CHECKS. WE DO NOT READ THEM AS POPULATION SUCCESS RATES.

Size	Hardware record	Method(s)	Runs	Recorded outcome
M8/M10/M12	registered conical scans	conical acquisition	26/size	78/78 valid traces
M12	fixed-seed method runs	PGS-O, PGS-G, Outward, Direct-4C	1/method	three terminal; Outward terminal after extension
M10	fixed-start pilots	PGS-G	2	both reached terminal depth

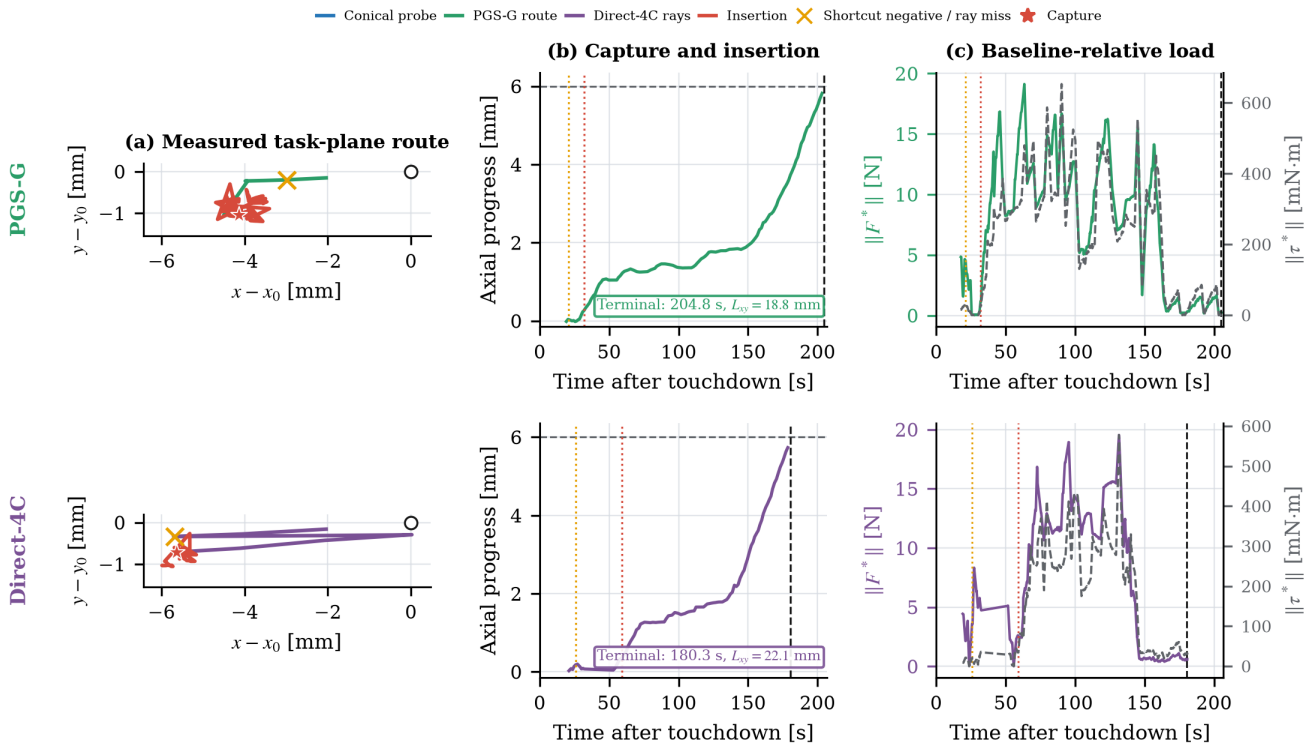


Fig. 9. Hardware mechanism comparison, two M12 runs from the same starting point. Rows: PGS-G, Direct-4C. (a) Measured task-plane route from each touchdown. Circles, crosses, and stars mark touchdown, negative evidence, and confirmed capture. (b) Axial progress, with the 6 mm task line. (c) Norms of baseline-relative force and torque at the Bota origin. Orange dotted, red dotted, and black dashed lines mark negative evidence, capture, and terminal events. Curves are raw, unsmoothed records. This pair is meant to show mechanism. It was not run as a randomized comparison.

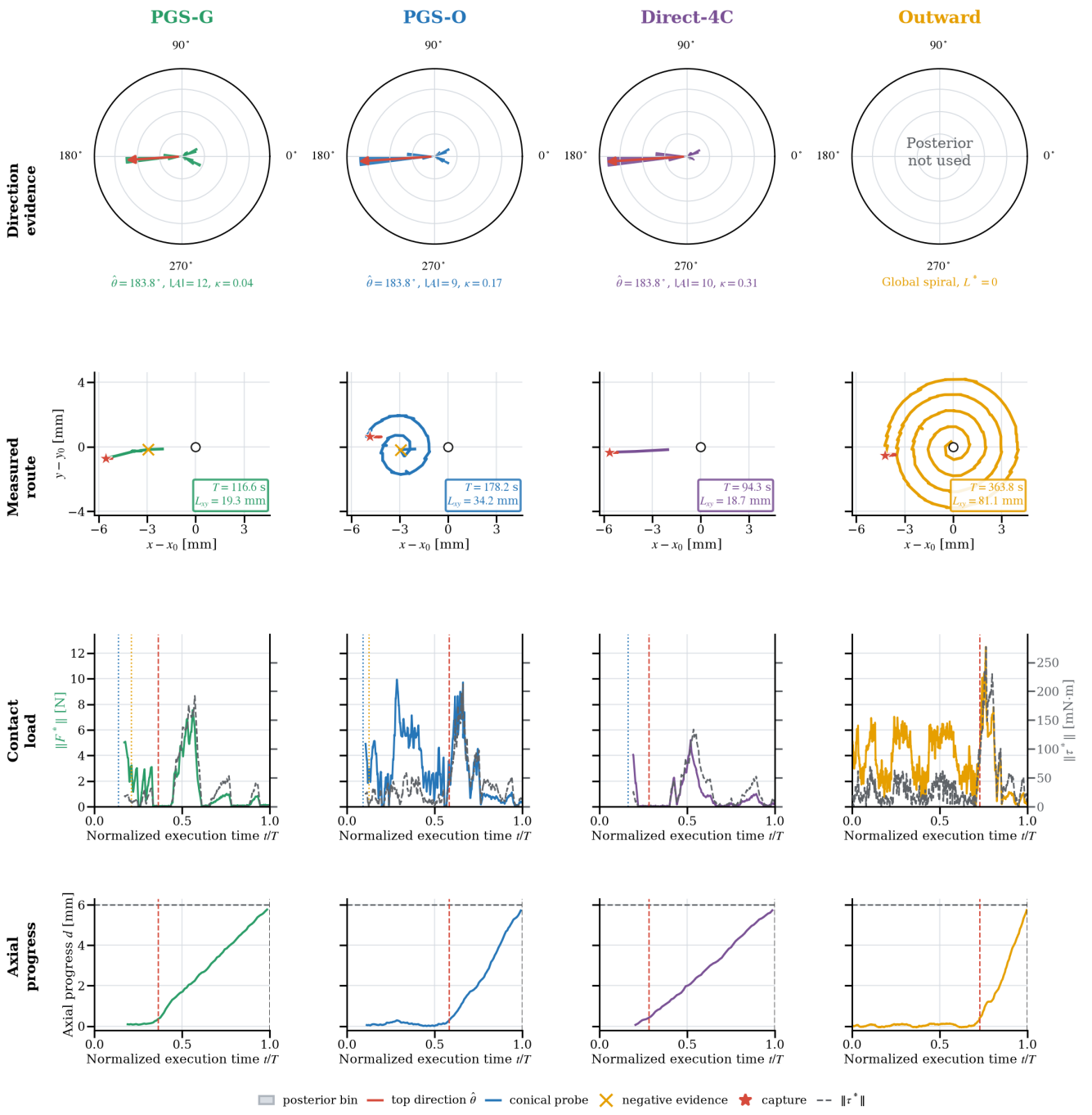


Fig. 10. Perception–action–contact matrix, one fixed-seed M12 hardware task. Columns: PGS-G, PGS-O, Direct-4C, Outward. Top row: the measured 48-bin direction posterior. Colored bins mark the prediction set. The red arrow marks the top direction. Outward has no tactile posterior at all. Second row: the measured task-plane route on shared axes. The circle, orange cross, and red star mark touchdown, valid negative evidence, and confirmed capture. Badges give terminal time and travel for that run. Third row: force (solid, left scale) and torque (gray dashed, right scale), both baseline-relative, over normalized time. Blue dotted, orange dotted, and red dashed lines mark probe end, negative evidence, and capture. Bottom row: axial progress from touchdown. The gray line is the 6 mm task datum. The black line is the terminal event. Every trace here is a raw, unsmoothed hardware record. This one campaign was picked to show mechanism and how effort gets spent, not to serve as a randomized comparison.

V. Discussion

The tactile scan turns out to play a narrow role. It changes where the search budget gets spent. It does not point straight at the socket center. That is why PGS-O routes stay short next to global outward search. It is also why the entropy grid barely helps here, since the residual regions tested are small to begin with. Recovery solves a different problem: it puts the robot back into a searchable contact state once an insertion try is rejected.

The gap between simulation and reality shows up mainly in the observation model. A model trained on the real tool sees friction, compliance, sensor chain, and registration errors that contact rendering leaves out or simplifies. It does not fix session drift, wear, temperature effects, or shifts across socket size. That is why the learned output stays a route proposal. Capture is still decided from the measured contact, not from the model.

A few limits are worth stating. Proposition 1 covers the retained mass of the empirical posterior under exact endpoint execution. It gives no guarantee once the distribution shifts. Boundary truncation, tracking error, or a budget that runs out can also stop a finite spiral or grid from covering its full set. The hardware records come from one fixture and a set of chosen starts, so we do not infer a random-start success rate from them. Bota readings are relative to a same-run baseline at the sensor origin, with no certified gravity compensation and no shift of moment to the tool tip. Last, the terminal dwell only confirms geometric engagement. It says nothing about first-turn coupling, slip, thread release, or extraction.

VI. Conclusion

This work treats tactile direction as partial evidence about the socket pose. It is not a full estimate of that pose. A conical probe updates the planar center belief. The confident part of that belief becomes a capped shortcut. What is left goes to a finite local search. Capture, recovery, and the final check all run through the same compliant loop, on every route. On the paired M8 tasks, this setup keeps the same engagement rate while cutting the route short, compared with an outward spiral. The recovery study and the M10/M12 hardware records show that this same split, learned guidance on one side and physical checking on the other, carries over to the real robot. The claim made here stops at geometric engagement. Turning the screw, releasing the thread, and extracting it each need their own contact rules, still to come.

References

[1] I. Díaz, D. Borro, O. Iparraguirre, M. Eizaguirre, F. A. Ricardo, N. Muñoz, and J. J. Gil, "Robotic system for automated disassembly of electronic waste: Unscrewing," *Robotics and Computer-Integrated Manufacturing*, vol. 95, p. 103032, 2025.

[2] D. E. Whitney, "Quasi-static assembly of compliantly supported rigid parts," *Journal of Dynamic Systems, Measurement, and Control*, vol. 104, no. 1, pp. 65–77, 1982.

[3] W. Lian, T. Kelch, D. Holz, A. Norton, and S. Schaal, "Benchmarking off-the-shelf solutions to robotic assembly tasks," in *Proceedings of the 2021 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2021, pp. 1046–1053.

[4] N. Hogan, "Impedance control: An approach to manipulation: Part I—theory," *Journal of Dynamic Systems, Measurement, and Control*, vol. 107, no. 1, pp. 1–7, 1985.

[5] F. Wirnshofer, P. S. Schmitt, G. von Wichert, and W. Burgard, "Controlling contact-rich manipulation under partial observability," in *Proceedings of Robotics: Science and Systems XVI*, 2020.

[6] S. Chintalapudi, L. Kaelbling, and T. Lozano-Pérez, "Bi-level belief space search for compliant part mating under uncertainty," in *Algorithmic Foundations of Robotics XVI, Volume 2*. Springer Nature Switzerland, 2026, pp. 107–124.

[7] M. Noseworthy, B. Tang, B. Wen, A. Handa, C. Kessens, N. Roy, D. Fox, F. Ramos, Y. Narang, and I. Akinola, "FORGE: Force-guided exploration for robust contact-rich manipulation under uncertainty," *IEEE Robotics and Automation Letters*, vol. 10, no. 5, pp. 4436–4443, 2025.

[8] S. L. Cleac'h, T. A. Howell, S. Yang, C.-Y. Lee, J. Zhang, A. Bishop, M. Schwager, and Z. Manchester, "Fast contact-implicit model predictive control," *IEEE Transactions on Robotics*, vol. 40, pp. 1617–1629, 2024.

[9] A. Aydinoglu, A. Wei, W.-C. Huang, and M. Posa, "Consensus complementarity control for multicontact MPC," *IEEE Transactions on Robotics*, vol. 40, pp. 3879–3896, 2024.

[10] W. Yang and M. Posa, "Dynamic on-palm manipulation via controlled sliding," in *Proceedings of Robotics: Science and Systems XX*, 2024.

[11] J. Carpentier, Q. L. Lidec, and L. Montaut, "From compliant to rigid contact simulation: A unified and efficient approach," in *Proceedings of Robotics: Science and Systems XX*, 2024.

[12] B. P. Graesdal, S. Y. C. Chia, T. Marcucci, S. Morozov, A. Amice, P. A. Parrilo, and R. Tedrake, "Towards tight convex relaxations for contact-rich manipulation," in *Proceedings of Robotics: Science and Systems XX*, 2024.

[13] H. J. T. Suh, T. Pang, T. Zhao, and R. Tedrake, "Dexterous contact-rich manipulation via the contact trust region," *The International Journal of Robotics Research*, vol. 45, no. 9, pp. 1418–1454, 2026.

[14] Q. Li, O. Kroemer, Z. Su, F. F. Veiga, M. Kaboli, and H. J. Ritter, "A review of tactile information: Perception and action through touch," *IEEE Transactions on Robotics*, vol. 36, no. 6, pp. 1619–1634, 2020.

[15] S. Jin, X. Zhu, C. Wang, and M. Tomizuka, "Contact pose identification for peg-in-hole assembly under uncertainties," in *Proceedings of the 2021 American Control Conference (ACC)*, 2021, pp. 48–53.

[16] S. Kim and A. Rodriguez, "Active extrinsic contact sensing: Application to general peg-in-hole insertion," in *Proceedings of the 2022 IEEE International Conference on Robotics and Automation (ICRA)*, 2022.

[17] R. Calandra, A. Owens, M. Upadhyaya, W. Yuan, J. Lin, E. H. Adelson, and S. Levine, "The feeling of success: Does touch sensing help predict grasp outcomes?" in *Proceedings of the 1st Annual Conference on Robot Learning*, ser. Proceedings of Machine Learning Research, vol. 78. PMLR, 2017, pp. 314–323.

[18] M. A. Lee, Y. Zhu, K. Srinivasan, P. Shah, S. Savarese, L. Fei-Fei, A. Garg, and J. Bohg, "Making sense of vision and touch: Self-supervised learning of multimodal representations for contact-rich tasks," in *Proceedings of the 2019 International Conference on Robotics and Automation (ICRA)*, 2019, pp. 8943–8950.

[19] C. Higuera, A. Sharma, C. K. Bodduluri, T. Fan, P. Lancaster, M. Kalakrishnan, M. Kaess, B. Boots, M. Lambeta, T. Wu, and M. Mukadam, "Sparsh: Self-supervised touch representations for vision-based tactile sensing," in *Proceedings of the 8th Conference on Robot Learning*, ser. Proceedings of Machine Learning Research, vol. 270. PMLR, 2025, pp. 885–915.

[20] J. Zhao, Y. Ma, L. Wang, and E. Adelson, "Transferable tactile transformers for representation learning across diverse sensors and tasks," in *Proceedings of the 8th Conference on Robot Learning*, ser. Proceedings of Machine Learning Research, vol. 270. PMLR, 2025, pp. 3766–3779.

[21] B. Liang, W. Liang, and Y. Wu, "Tactile-guided dynamic object planar manipulation," in *Proceedings of the 2022 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2022, pp. 3203–3209.

[22] B. Ai, S. Tian, H. Shi, Y. Wang, C. Tan, Y. Li, and J. Wu, "RoboPack: Learning tactile-informed dynamics models for dense packing," in *Proceedings of Robotics: Science and Systems XX*, 2024.

[23] M. Oller, D. Berenson, and N. Fazeli, "Tactile-driven non-prehensile object manipulation via extrinsic contact mode control," in *Proceedings of Robotics: Science and Systems XX*, 2024.

[24] J. D. A. Ferrandis, J. Moura, and S. Vijayakumar, "Learning visuotactile estimation and control for non-prehensile manipulation under occlusions," in *Proceedings of the 8th Conference on Robot Learning*, ser. Proceedings of Machine Learning Research, vol. 270. PMLR, 2025, pp. 1501–1515.

[25] B. Huang, Y. Wang, X. Yang, Y. Luo, and Y. Li, "3D-ViTac: Learning fine-grained manipulation with visuo-tactile sensing," in *Proceedings of the 8th Conference on Robot Learning*, ser. Proceedings of Machine Learning Research, vol. 270. PMLR, 2025, pp. 2557–2578.

[26] K. Yu, Y. Han, Q. Wang, V. Saxena, D. Xu, and Y. Zhao, "MimicTouch: Leveraging multi-modal human tactile demonstrations for contact-rich manipulation," in *Proceedings of the 8th Conference on Robot Learning*, ser. Proceedings of Machine Learning Research, vol. 270. PMLR, 2025, pp. 4844–4865.

[27] H. Lin, R. Corcodel, and D. Zhao, "Generalize by touching: Tactile ensemble skill transfer for robotic furniture assembly," in *Proceedings of the 2024 IEEE International Conference on Robotics and Automation (ICRA)*, 2024, pp. 9227–9233.

[28] G. Schoettler, A. Nair, J. A. Ojea, S. Levine, and E. Solowjow, "Meta-reinforcement learning for robotic industrial insertion tasks," in *Proceedings of the 2020 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2020, pp. 9728–9735.

[29] T. Z. Zhao, J. Luo, O. Sushkov, R. Pevceviciute, N. Heess, J. Scholz, S. Schaal, and S. Levine, "Offline meta-reinforcement learning for industrial insertion," in *Proceedings of the 2022 IEEE International Conference on Robotics and Automation (ICRA)*, 2022, pp. 6386–6393.

[30] C. Chi, S. Feng, Y. Du, Z. Xu, E. Cousineau, B. Burchfiel, and S. Song, "Diffusion policy: Visuomotor policy learning via action diffusion," in *Proceedings of Robotics: Science and Systems XIX*, 2023.

[31] Y. Chen, K. Kimble, H. H. Qian, P. Chanrungmaneekul, R. Seney, and K. Hang, "Robust peg-in-hole assembly under uncertainties via compliant and interactive contact-rich manipulation," *arXiv preprint arXiv:2506.22766*, 2025. [Online]. Available: https://arxiv.org/abs/2506.22766

[32] J. Xue, W. Liang, Y. Xu, Y. Wu, and T. H. Lee, "Learning-based predictive impedance control towards safe predefined-time physical robotic interaction," in *Proceedings of the 2025 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2025, pp. 18 543–18 550.

[33] Y. Xu, W. Liang, J. Xue, Y. Wu, and T. H. Lee, "Sequen-sync contact force/torque control using nested fast terminal sliding mode control approach," in *Proceedings of the 2025 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2025, pp. 6876–6883.

[34] Y. Romano, M. Sesia, and E. J. Candès, "Classification with valid and adaptive coverage," in *Advances in Neural Information Processing Systems*, vol. 33, 2020, pp. 3581–3591.